UNIVERSITY OF CALIFORNIA SAN DIEGO

Increasing the Trigger and Selection Efficiency for Higgs to Diphoton Signal at the CMS Experiment Using Machine Learning Techniques

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in

Physics

by

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ABSTRACT OF THE DISSERTATION

Increasing the Trigger and Selection Efficiency for Higgs to Diphoton Signal at the CMS Experiment Using Machine Learning Techniques

by

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James G. Branson, Chair

The CMS and ATLAS experiments at the LHC announced the discovery of the Higgs boson in 2013, 50 years after it was originally predicted to resolve issues in earlier formulations of the Standard Model (SM) of particle physics. Since announcing its discovery, both CMS and ATLAS have undertaken progressively more precise measurements of the properties of the Higgs boson. The high data rate of the LHC is reduced by a two-tiered triggering system. The software-based High Level Trigger (HLT) must process events and reduce this rate from 40 MHz to just 7 kHz of proton bunch crossings. One particularly interesting channel, $H \to \gamma\gamma$, has been instrumental in both the discovery and later refinements to the measurements of the Higgs

boson. In Run 2, the $H \to \gamma\gamma$ HLT was based on hard cuts on kinematic, shower shape, and isolation variables. This paper presents a new $H \to \gamma\gamma$ HLT based on a Boosted Decision Tree (BDT) machine learning model. This new trigger is able to achieve 97.32% signal efficiency on $H \to \gamma\gamma$ MC samples. Furthermore, the overall data rate is reduced from 61.53 Hz to 29.43 Hz by the new trigger. A complementary offline selection, also based entirely on new BDTs, has been developed. This new selection allows a 29.4% gain in signal MC efficiency relative to the Run 2 standard analysis cuts (91.27% vs 70.53%). Results will additionally be shown for the rare $HH \to b\bar{b}\gamma\gamma$ DiHiggs decay channel, where a trigger-level efficiency of 98.45% and a corresponding offline selection efficiency gain of 39.7% is achieved.

Introduction

First theorized in 1964, the Higgs field consitutes a vital component of the Standard Model (SM) of particle physics. The insertion of this field to the SM allowed a mechanism for the spontaneous breaking of the electroweak symmetry, a necessary fix to the versions of the antecedent theories of fundamental particles. This spontaneous symmetry breaking provides an avenue though which the fundamental particles may gain their masses; until the formulation of the Higgs fields, the prevailing version of the SM could not account for the existence of – let alone the values of – the masses of quarks and the force-carrying gauge bosons. Luckily, the prediction of the Higgs field proposed a corresponding bosonic particle to mediate the interactions of this field. As the fundamental particles prediced by the SM were discovered throughout the 20th century, the Higgs boson proved to be elusive, in fact eventually becoming the last undiscovered fundamental particle predided by the SM.

Since the discovery of the Higgs boson in 2012, the Large Hadron Collider (LHC) has undertaken continuously more precise measurements of its properties. As the amount of available data has accumulated, increasingly rare channels of analysis have become explorable. To enable these high-quality analyses, the hardware and software of the LHC and its detectors has had to be improved considerably. Due to the abundance of background processes created at the high energies achieved by the LHC, coupled with the relative rarity of the processes by which the Higgs boson is produced and decays, complex software is necessary to disentagle the signal processes.

This software is constructed using a layered approach, where each subsequent step in the processing of the data serves to further reduce the total ammount of data events being analyzed,

while, ideally, preserving as many events considered signal as possible. The first stages of this data reduction are achieved by a set of triggers. These triggers are relatively coarse-grain, seeking to reject the most obvious background events and thereby reduce the amount of data being stored and analyzed by three to four orders of magnitude. Of course, given the large number of different final states of the Higgs boson decay, many different triggers must be developed to dig out the desired signal signature required for a specific analysis. Once these triggers are applied, further analysis software is used on the resulting data to produce final results of the measurements of the various properies of the desired physics process.

The goal of this paper is to examine the improvement of a specific set of these triggers and analysis-level cuts. While many channels of Higgs boson decay exist, the decay into two photons has been proven to be a major component of the analysis of the Higgs boson; this process provides a relatively clean decay signature, and occurs at a high enough rate to provide ample statistics for analysis. Additionally, the diphoton channel can play a vital role in the search for diHiggs production. Lower-rate processes, such as this diHiggs production, benefit greatly from improved signal efficiency, so accurate triggering is a vital component of probing both SM and beyond-SM physics.

While the historical standard for triggering at the Compact Muon Solenoid (CMS) experiment was to place carefully selected cuts on various measureable variables, the use of machine learning (ML) techniques has exploded over the last 10 years. Since triggering is a binary classification problem, ML techniques are exceedingly well-suited to perform the necessary selections. This paper will explore the development and application of a ML-based approach to triggering for Higgs to two photons processes at CMS. A theoretical description of the SM and Higgs Boson will be given in chapter 1. The CMS detector and the detector elements relevant to the development of the $H \to \gamma\gamma$ trigger will be given in chapter 2. The structure of CMS triggers and the software used to run and evaluate them will be discussed in chapter 3. Chapter 4 will provide details on the Boosted Decision Tree ML technique, as well as the specific software used to develop one for the CMS triggering system. Chapters 5 and 6 will describe the

development of the BDT-based trigger and a complementary downstream selection for $H \to \gamma\gamma$. Finally, chapter 7 will present the improvement in signal efficiency and data rate provided by this novel triggering approach.

Chapter 1

The Standard Model and Higgs Boson

The standard model (SM) of particle physics is the quantum gauge field theory that describes all currently known physical interactions besides gravity [1]. Early formulations considered there to be three distinct gauge theories that ruled interactions between physical objects: Quantum Electrodynamics (QED), which describes the elctromagnetic force between charged objects; Quantum Chromodynamics (QCD), which describes the strong nuclear force; and Weak Theory, which describes the weak nuclear force.

Within the SM, there are two major forms of particles. Spin 1/2 particles, called fermions, comprise all matter as we know it, and include quarks and leptons. Integer spin particles, called bosons, act as the force carriers that mediate interactions between these fermions. QED posits the photon as the gague boson, which mediates interactions between all electrically charged particles – including both charged leptons and quarks. This theory will be touched on in section 1.1.1. QCD describes eight gluons as the force mediators between color-charged quarks. A minimal summary of QCD will be given in section 1.1.2. The weak theory describes the massive $W^{\pm}$ and Z^0 bosons as the mediators of the weak force, which is responsible for radioactive decay. More details on this theory, as well as its unification with QED, will be given in section 1.1.3.

The final piece of the standard model, the Higgs Boson, was first theorized in 1964 in a seminal paper by Peter Higgs [2]. This addition to the SM was motivated by the need for a way to unify QED with weak theory, as well as problems with the theories of massive gauge

bosons, as will be discussed in section 1.2. The Higgs boson wasn't discovered until the LHC released its first results in 2012; previous accelerators were not capable of producing the Higgs boson in the necessary quantities to be studied at a mass of 125 GeV/c^2. These results, published simultaneously by the CMS and ATLAS experiments at LHC, established the existence of the Higgs Boson and provided the final piece to the standard model [3].

In figure 1.1, an overview of the component fundamental particles of the SM is presented. The three generations of quarks are shown in purple. These quarks are bound together by the strong nuclear force, which is mediated by gluons and described by QCD. Below the quarks in green are the three generations of leptons, which interact through the electroweak force mediated by the photon, $W^{\pm}$, and Z^0 bosons. All four of these force-mediators, termed gauge bosons, are shown in red. Finally, the Higgs boson, which occupies a special role in electroweak theory, is shown in yellow.

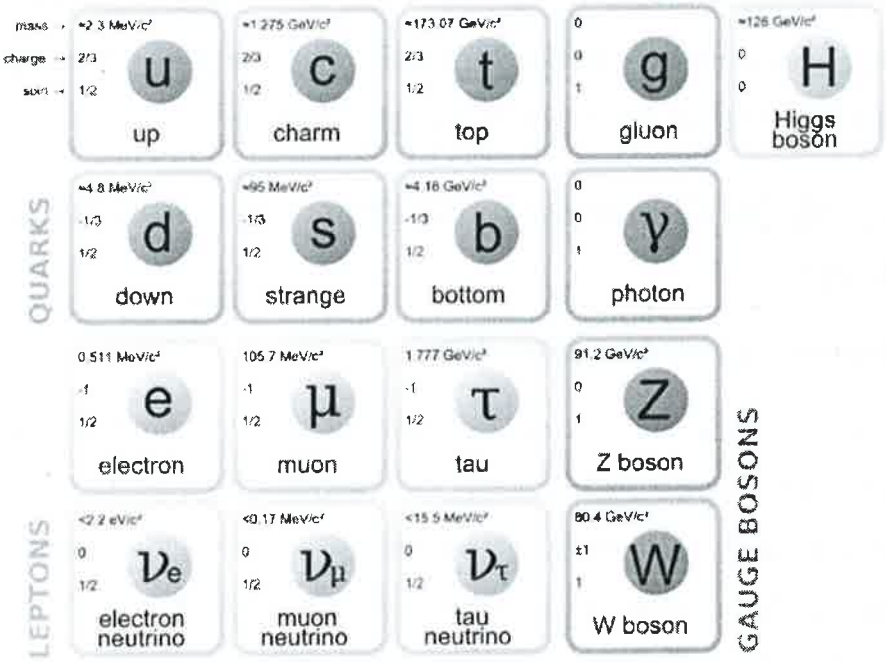


Figure 1.1. A chart of all the known irreducible particles of the standard model of particle physics. The three generations of quarks are shown in purple. The three generations of leptons are shown in green. The gauge bosons, which mediate the strong, weak, and electromagentic forces are shown in red. The Higgs boson is shown in yellow.

In rest of this chapter, Feynman diagrams will be used to represent particle interactions. These diagrams are schematics showing the quanta involved in an interaction (represented as straight, wavy, or curly lines) and intersection points where the interactions occur. By general convention, moving from left to right in such a diagram corresponds to going forward in time. For particles with a corresponding antiparticle (fermions), an arrow on a line pointing towards the right generally denotes the particle while an arrow pointing towards the left denotes the antiparticle. Generally, quarks and leptons are shown as solid lines, bosons such as the photon, W, and Z are shown as wavy lines, gluons are shown as curly lines, and a dashed line is used to represent the Higgs boson.

In order to calculate the amplitudes of an interaction in particle physics, complicated path integrals must be computed. Feynman diagrams introduce so-called Feynman rules, which allow these integrals to be constructed using the diagrams themselves [4]. As a reuslt, Feynman diagrams are an invaluable tool to not only qualitatively understand how particles interact in QFTs but also to construct and calculate the relevant amplitudes necessary for a theoretical understanding of the production and decay of fundamental particles.

1.1 The Major Quantum Field Theories of the Standard Model

1.1.1 Quantum Electrodynamics (QED)

Quantum electrodynamics was the first successful field theory proposed, and was originally created to describe the phenomena of the electromagnetic force [1]. This original theory was built on U(1) gauge symmetry, and thererefore has one generator corresponding to one gauge boson, the photon. In this theory, there is one type of charge: electric charge can either be positive or negative. Interactions in QED occur when a photon couples with any electrically charged fundamental particle.

1.1.2 Quantum Chromodynamics (QCD)

Another major field theory developed as part of the SM is Quantum Chromodynamics, or QCD. This theory governs the interactions of quarks through the strong nuclear force. QED hinges on the existence of electric charge which can be, of course, either positive or negative. QCD, instead, has an associated quantum number termed "color charge". Quarks can carry one of three color (red, blue, green), whereas anti-quarks carry the same anti-color charges (denoted as $\bar{r}$, $\bar{b}$, and $\bar{g}$). All bound states governed by the strong force, known as hadrons, are color-neutral. Mesons, the lightest hadronic particles, have a quark with a color charge and an anti-quark with an anti-color charge, forming a "color singlet". Baryons, which include the proton and neutron, contain all three color charges which also form a color-singlet combination (similar to how a proton and electron together form an electrically-neutral combination).

Just as QED is mediated by massless photons, the strong force is mediated by massless particles called gluons. Unlike the photon, which is electrically-neutral, gluons carry color charge. To mediate between a red quark and anti-red anti-quark, for instance, a gluon must contain a $r\bar{r}$ combination. One might immediately imagine that there would be 9 gluons, then, corresponding to the 3x3 combination of all possible color-anti-color pairs. The underlying field theory of QCD, however, is of the SU(3) symmetry group. This group only has 8 generators, which correspond to 8 possible gluon combinations [5].

One vital component of QCD is quark confinement. In QED, the electromagnetic force falls off with increasing distance; this fact allows nearby charges to interact, but increasing the separation between the charges is able to weaken the force until the two objects are no longer bound. In contrast, the strong nuclear force actually increases in magnitude as the separation between two color-charged particles increases [1]. So, as two quarks are separated from eachother, the bond between them contains more and more energy. As the separation grows larger, there comes a point where the energy stored in the gluons binding the quarks together is larger than the energy required to produce a new quark-antiquark pair.

Quark confinement leads to the production of what are termed QCD jets [5]. At the high collision energies achieved at the LHC, enough energy is available to create many new quark-antiquark pairs from a single interaction. This leads to a cascading effect, where newly-created quarks may form particles momentarily – a process called hadronization – before separating and creating even more new quarks. This cascade can produce many new hadrons, which have momentum in approximately the same direction. These jets of hadrons occur at very high rates, and constitute both a source of signal and a major form of background for many analyses conducted at the LHC.

1.1.3 Electroweak (EW) Theory

The weak nuclear force was originally formulated by Enrico Fermi in 1938 as a way to explain the process of beta decay of nuclei [6]. While this force was originally proposed as the 4th fundamental force, it was also not a complete description at the time of its formulation. Namely, the original formulation was not renormalizable, which essentially means that the calculations stemming from the theory sometimes diverge and yeild infinities. After QED began to prove highly fruitful, great efforts to formulate the weak interaction as a renormalizable field theory were undertaken in earnest in the 1950s. Then, between 1959 and 1961, papers by first Salam and Ward, and then Glashow, proposed a major step forward in formulating the weak force as a field theory [7] [8]. This breakthrough came in the form of unifying the electromagnetic and weak nuclear forces into one single electroweak theory.

By combining the SU(2) gauge symmetry of the weak force with the U(1) symmetry of QED, a new group with total symmetry $SU(2)_L \times U(1)$ can be formulated [5]. This correctly predicts the existence of 3 gauge bosons corresponding to the weak part of the symmetry (called the charged $W^{\pm}$ and neutral Z^0 boson), and one gauge boson corresponding to the electromagnetic term (the photon). Requiring gauge invariance of this theory, however, implies that the 3 weak force gauge bosons must be massless. Due to the limited interaction range of the weak force, however, it was clear that the gauge bosons responsible for the weak force must have some finite

(and relatively large) mass. If a mechanism was introduced that broke the SU(2) symmetry, however, electroweak theory could allow for massive W and Z bosons [1].

The concept of spontaneous symmetry breaking was first understood in terms of quantum field theories by considering a lagrangian for a simple scalar field that obeys U(1) symmetry. The general Lagrangian for such a field can be given by equation 1.1, where ϕ represents the field itself, ∂^μ is the partial derivartive relative to a spacetime dimension, μ, and $V(\phi)$ is some general potential.

$$\mathscr{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi) \tag{1.1}$$

The first term in this equation is invariant under rotations of the form $\phi \rightarrow e^{-i\alpha}\phi$, and represents the kinetic term of the Lagrangian. The second term, the potential term, is where this U(1) symmetry can be broken. Then, a potential of the form shown in equation 1.2 is introduced to the lagrangian [9]. In this potential, λ and η are taken to be constants. This is still invariant under global phase translations of the form $\phi \rightarrow e^{-i\alpha}\phi$.

$$V(\phi) = \frac{1}{2}\lambda(\phi^*\phi - \frac{1}{2}\eta^2)^2 \tag{1.2}$$

The key feature of this potential, however, is that the field achieves a local maximum at $\phi = 0$. This is termed as a non-zero vacuum expectation value (VEV). Instead, the lowest energy state (ground) state can be seen in equation 1.3.

$$\langle 0|\phi(x)|0\rangle = \frac{1}{\sqrt{2}}\eta e^{i\alpha} \tag{1.3}$$

This represents a vacuum state which is infinitely-degenerate in the coordinate α. The VEV being non-zero means that the symmetery respected by the initial lagrangian is not respected by the vacuum state of the field. The ground (lowest energy) state, however, is now removed from $\phi = 0$. This leads to the so-called "sombrero potential", given that there is a peak at 0 then

a lower state removed from the axis as seen in figure 1.2 (reproduced from [10])

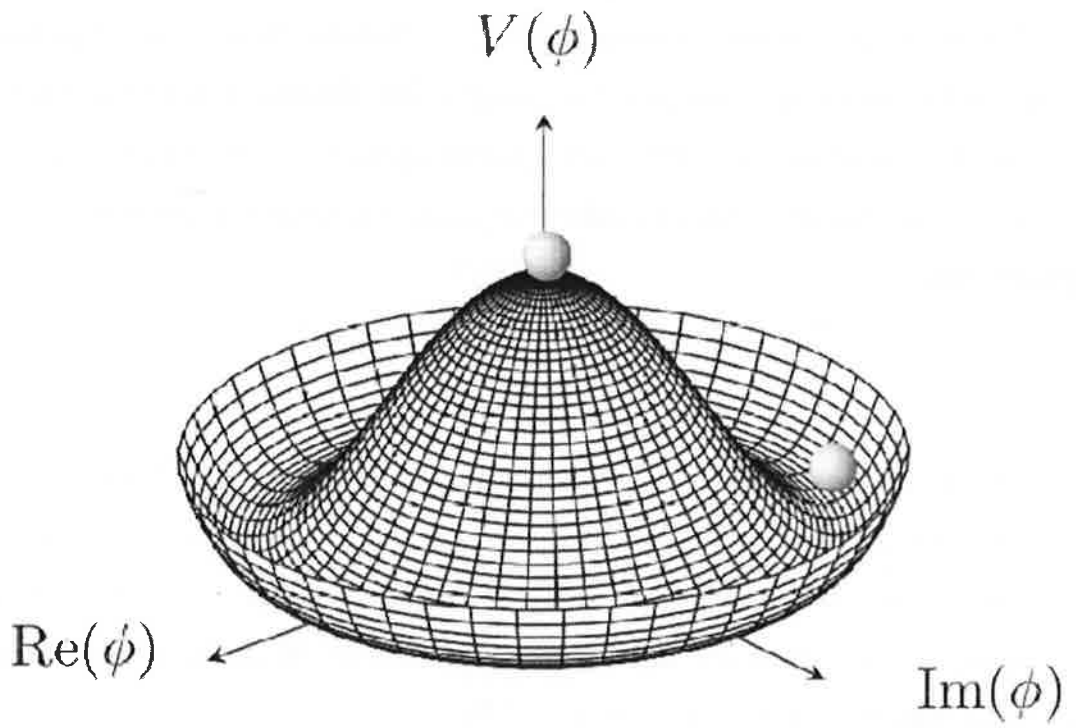


Figure 1.2. The "Sombrero Potential" which spontaneously breaks the symmetry of an example U(1) group (reproduced from [10])

This mechanism, originally proposed by Yoichiro Nambu in the context of superconductivity and expanded upon by Jefferey Goldstone in the early 1960s, made a prediction that became a huge sticking point in the development of the electroweak theory; while it allows a mechanism for spontaneous symmetry breaking, Goldstone's theorem predicts the existence of a massless spin-0 Nambu-Goldstone boson [9]. These bosons should be easy to detect, but had never been seen. So, an additional mechanism was necessary to explain the absence of these Nambu-Goldstone bosons.

1.2 The Brout–Englert–Higgs Field and Higgs Boson

1.2.1 The Brout–Englert–Higgs (BEH) Mechanism

The final piece of electroweak unification that allowed both spontaneous symmetry breaking and the lack of Nambu-Goldstone bosons was elucidated separately in 1964 by Brout and Englert [11], Higgs [2], and Guralnik, Hagen and Kibble [12]. This mechanism, later termed the Brout-Englert-Higgs (BEH) mechanism, adds the kinetic term for a gauge field, A_μ, into the lagrangian. Then, the lagrangian of the Goldstone formulation can further be written in a gauage-invariant way by replacing the partial derivative with the covariant derivated as given in equation 1.4. This is termed the Abelian Higgs model, as it is the Lagrangian for an Abelian gauge group. In this formulation, the covariant derivative is defined as given in equation 1.5 and a field tensor is introduced as defined in equation 1.6 [9]. In this case, g is the fundamental unit charge.

$$\mathscr{L} = D_\mu \phi^* D^\mu \phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2}\lambda(\phi^*\phi - \frac{1}{2}\eta^2)^2 \tag{1.4}$$

$$D_\mu = \partial_\mu - igA_\mu \tag{1.5}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1.6}$$

For this formulation, it becomes important to decompose the scalar field ϕ into real and imaginary components as given in equation 1.7. Here, φ_1 will be the real component of the complex scalar field, ϕ, whereas φ_2 will be the imaginary component.

$$\phi = \frac{1}{\sqrt{2}}(\eta + \varphi_1 + i\varphi_2) \tag{1.7}$$

Now, one last convenient choice can be made to make the final result more illuminating. A new vector field can be defined in terms of A_μ as given in equation 1.8.

$$B_\mu = A_\mu + \frac{1}{g\eta}\partial_\mu \varphi_2 \tag{1.8}$$

This takes the form of a gauge transformation, which allows the field tensor to remain unchanged with $A_\mu \to B_\mu$. The final step is to insert the definitions of equations 1.5, 1.7, and 1.8 into the lagrangian defined above. While this will lead to many terms, it turns out that all of the important information that must be gleaned from this mechanism can be seen while discarding terms above those quadratic in φ_1. So, then, the final result can be written as in equation 1.9.

$$\mathscr{L} = \frac{1}{2}\partial_\mu \varphi_1 \partial^\mu \varphi_1 - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}\lambda\eta^2\varphi_1^2 - \frac{1}{2}g^2\eta^2 B_\mu B^\mu. \tag{1.9}$$

It can immediately be seen that there are two kinetic terms in this equation: $\frac{1}{2}\partial_\mu \varphi_1 \partial^\mu \varphi_1$ and $-\frac{1}{2}g^2\eta^2 B_\mu B^\mu$. So then, there are no more massless terms. Instead, with the deployment of the BEH mechanism, there is a massive scalar, φ_1, and a massive vector, B_μ. Therefore, this mechanism would allow for spontaneous symmetery breaking without the need for massless bosons. Of course, this is still essentially a toy version of the theory that introduces the BEH mechanism as a way to produce the necessary properties beginning from the goldstone lagrangian. Work still needed to be done to extend this mechanism into the SM as a way to unify electromagnetism and the weak force in a way that was consistent with existing measurements [1].

1.2.2 The BEH Mechanism and Electroweak Unification

With the tool of BEH created, the electroweak gauge theory could finally be constructed in a way consistent with observations of previous experiments. In 1967, Weinberg and Salam built on the work Glashow had done 6 years prior by inserting the BEH spontaneous symmetry breaking into the existing attempts at electroweak unification [13]. The electroweak gauge theory is invariant under the $SU(2)_L \times U(1)_Y$ symmetry group, where the L signifies the doublets of left-handed chirality, and Y signifies the weak hypercharge scalar [5].

Just as a field with a potential given in equation 1.4 can be inserted into the gauged version of the Goldstone theory, a field can be inserted into the electroweak lagrangian to spontaneously break the symmetry of the $SU(2)_L \times U(1)_Y$ group. This field is usually written in the form given by equation 1.10.

$$V(\Phi) = m^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2 \tag{1.10}$$

One key difference of this formulation is that now the Higgs field, Φ, is a complex doublet in $SU(2)_L$, meaning that is has 4 degrees of freedom. In the definition given in equation 1.11, ϕ^0 and a^0 are neutral components that are even and odd in charge conjugation and parity (CP) symmetry, respectively. ϕ^+, on the other hand, is the complex charged component.

$$\Phi = \frac{1}{\sqrt{2}}\begin{bmatrix} \sqrt{2}\phi^+ \\ \phi^0 + ia^0 \end{bmatrix} \tag{1.11}$$

If λ is taken as negative in equation 1.10, then the potential takes the form of a sombrero, as in the original formulation of electroweak symmetry breaking. Then, the vacuum expectation value can again be taken, yielding the form givne in equation 1.12.

$$\langle 0|\Phi|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ v \end{bmatrix} \tag{1.12}$$

The value, v, is termed the Higgs vacuum expectation value, and is found to be $v = 246.22$ GeV [5]. Three of the four generators of the $SU(2)_L \times U(1)_Y$ symmetry are then spontaneously broken, implying the existence of 3 massless goldstone bosons. Of course, given the previous discussion of the BEH mechanism, this can be resolved. A new Higgs lagrangian can then be written with the covariant derivative, as before. This is shown in equation 1.13

$$\mathscr{L} = (D_\mu\Phi)^\dagger(D^\mu\Phi) - m^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2 \tag{1.13}$$

Now, the covariant derivative must be defined in terms of the gauge fields defined for the $SU(2)_L \times U(1)_Y$ group. Whereas the original formulation of BEH was constructed with respect to the gauge field A_μ, the electoweak lagrangian must be written in terms of the W_μ and B_μ gauge fields of the weak and electromagnetic fields. Just as there was a coupling constant associated with the coupling of the generic BEH field to the goldstone lagrangian, there are coupling constants associated with the coupling of the Higgs field to the W_μ and B_μ fields. These constants, g (for the $SU(2)_L$ gauge) and g' (for the $U(1)_Y$ gauge) are inserted into the covariant derivative along with the corresponding fields as given in equation 1.14 [5].

$$D_\mu \Phi = (\partial_\mu + 1/2(ig\sigma^a W_\mu^a/2 + ig' Y B_\mu))\Phi \tag{1.14}$$

Here, the index a = 1,2,3 runs over the 3 usual pauli matrices, σ^a. So, then, the covariant derivative is now a matrix that "mixes" the three goldstone bosons generated by breaking the $SU(2)_L$ symmetry with the Higgs field, giving them masses. The fourth generator of the $SU(2)_L \times U(1)_Y$ symmetry is not broken, and therefore remains massless. This unbroken generator corresponding to the U(1) component of the gauge is the massless photon. The masses of the W and Z bosons, corresponding to the charged and uncharged massive gauge bosons of the electroweak force, can be given in terms of the coupling constants as in equation 1.15 [5].

$$m_W^2 = \frac{g^2 v^2}{4} \qquad\qquad m_Z^2 = \frac{(g'^2 + g^2)v^2}{4} \tag{1.15}$$

These masses, then, depend on the coupling parameters. In a practical sense, the logic of that is best reversed; by measuring the masses of the W and Z bosons, the coupling parameters g and g' can be measured. Many experiments have been undertaken to measure the masses of these vector bosons, and the particle data group has published a value for each as given in equation 1.16 [5]. These values are based on a combination of all of the measurements obtained by different groups and different experimental methods.

$$m_W = 80.3692 \pm 0.0133\ GeV/c^2 \qquad m_Z = 91.1880 \pm 0.0020\ GeV/c^2 \tag{1.16}$$

1.2.3 Properties of the Higgs Field and the Higgs Boson

So, then, of the four degrees of freedom in the complex Higgs scalar doublet, Φ, three have been "consumed" to generate the masses for the $W^{\pm}$ and Z^0 vector (spin 1) gauge bosons. The remaining degree of freedom manifests as a new scalar (spin 0) gauge boson called the Higgs boson. This Higgs boson can be seen as the manifestation of the Higgs field, and provides a measureable particle with which to understand the underlying Higgs field. The Higgs boson has neutral electric charge, and does not couple to either the photon or gluon [1].

As with the W and Z bosons, the mass of the Higgs boson is considered a parameter in the standard model. By constructing the lagrangian for the interaction between the Higgs and vector bosons, a mass in terms of previously-discussed quantities can be obtained as in equation 1.17. Here, v is the Higgs vacuum expectation value and λ is the constant introduced within the Higgs potential. The vev can be measured by its relation to the fermi constant (which can be probed by muon decay rate), or the masses of the vector bosons. Current measurements put this vev at $v = 246.22$ GeV [5].

$$m_H = \lambda v \tag{1.17}$$

There are still two parameters in this relationship that cannot be theoretically derived or measured by other methods. This means that there was no theoretical prediction whatsoever for the mass of the Higgs boson before its discovery. While some broad limits on the range of the mass could be calculated from radiative corrections to the SM, the Higgs boson mass was still a relatively free parameter of the theory. If the mass of the Higgs boson can be measured, however, the parameter λ can be inferred and the overall shape of the Higgs potential can be

better understood. Since this potential is responsible for the masses of the massive vector bosons, as well as all other fundamental particles (through the Yukawa interaction), measurement of the Higgs boson mass is of vital importance [13].

The BEH potential can be rewritten in a way that aligns more with the phenomenological measurements that can be taken at particle accelerators. Equation 1.18 shows the potential in terms of the Higgs mass, m_H, the parameter λ, and powers of the Higgs field. Each of these terms can be used to better understand the Higgs field in terms of particle interactions. The first term immediately accounts for the mass of the Higgs boson. The second term, however, shows the meaning behind the parameter λ. The factor Φ^3 in this term is called a tri-linear coupling; this represents an interaction vertex between three Higgs bosons. So, then, the parameter λ can be understood as the self-coupling strength of the Higgs. Whereas many particles in the standard model do not couple to themseleves, the Higgs does. This has far-reaching implications for the allowed production and decay modes of the Higgs boson, as will be discussed further in section 1.3.

$$V(\Phi) = \frac{1}{2}m_H^2\Phi^2 + \sqrt{\lambda/2}m_H\Phi^3 + \frac{1}{4}\lambda\Phi^4 \tag{1.18}$$

1.3 Phenomenology of the Higgs Boson

1.3.1 Production Modes of a Single Higgs Boson

With the establishment of the existence of the Higgs boson, both the production and decay modes of the particle must be studied and measured. The production modes are all of the theoretically-allowable ways in which a Higgs boson may be created. Determining which modes are allowable is a complicated process of using the relevant quantum field theories (QCD and EW theory for the Higgs boson) to constrain possible interactions using the conservation laws of each theory.

The primary production modes for the Higgs boson are all shown as feynman diagrams

in figure 1.3, reproduced from [13]. Figure 1.3a shows what is called gluon fusion (ggH), where two gluons directly fuse through a virtual (top or less commonly bottom) quark loop. It can immediately be seen that this ggH production mode does not produce any additional particles from the primary interaction. The next mode, in figure 1.3b is known as Vector Boson Fusion (VBF). This occurs when two quarks each radiate a vector boson (either $W^{\pm}$ or Z^{0}). These vector bosons then interact, and can produce a Higgs boson. VBF production also only produces a single Higgs boson, with the other two outgoing particles simply being the recoiling quarks that originally produced the vector bosons.

The other Higgs production modes shown are called associated production modes; instead of simply combining the energy of two particles together to produce a single Higgs boson, associated production modes create the Higgs in conjunction with some other particle. In figure 1.3c, vector boson associated production is shown. In this mode, two quarks directly interact to create a vector boson, which then branches into a Higgs in association with an additional vector boson. For this process, that vector boson may be either a $W^{\pm}$ or Z^{0}. The next diagram, shown in figure 1.3d, is the production of the Higgs in association with a quark pair, either the top or bottom quark. These are termed ttH or bbH production, and often produce additional jets of hadronic particles that originate from the top or bottom quarks. While these additional jets are useful for correctly tagging an event as ttH or bbH, they may also prove hard to measure accurately and therefore introduce considerable uncertainty to the measurements of these processes. The final two diagrams, 1.3e&f, are the production of a Higgs in association with a single top quark.

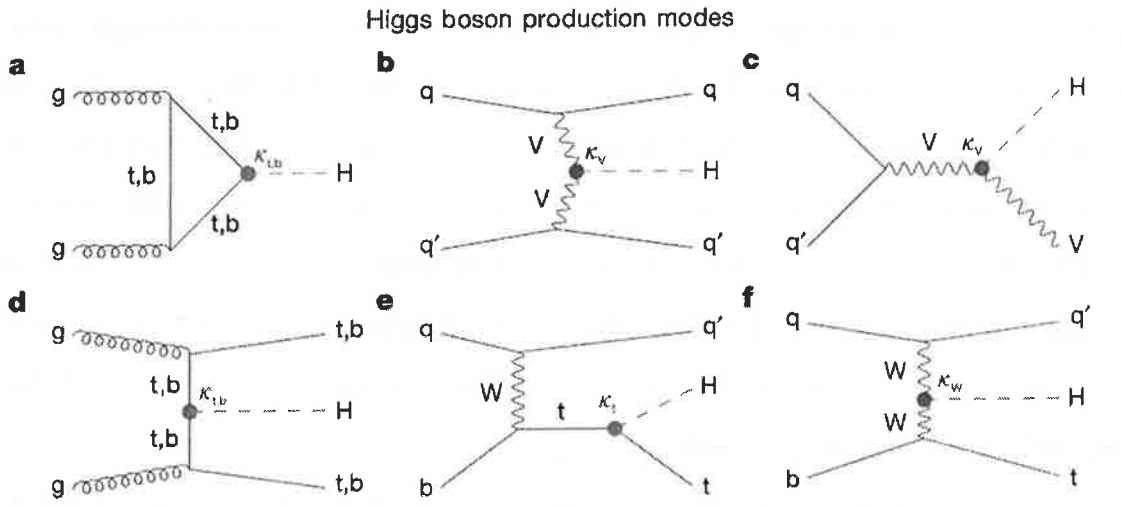


Figure 1.3. Feynman diagrams of known Higgs production modes. **a** is known as gluon fusion, termed ggH. **b** is known as vector boson fusion (VBF). **c** is production of the Higgs in association with a vector boson (either a $W^{\pm}$ or Z^0 boson), called VH production. **d** is the production of the Higgs in association with a pair of top (ttH) or bottom (bbH) quarks. **e** and **f** both show Higgs production in association with a single top quark (tH). Reproduced from [13]

Of note are the factors placed at each of the vertices in these diagrams. Coupling parameters are part of the field-theoretical matrix element calculations, and are a factor placed at each vertex of a particular particle combination. In experimental measurements, however, the coupling parameters are generally measured as coupling modifiers, denoted by κ, which are the ratio of the measured coupling parameter to the SM prediction for its value [13]. For the coupling of a Higgs and a top quark, for example, a value of $\kappa_t = 1$ would mean that the measured coupling parameter exactly matches that predicted by the standard model. The measurement of the coupling parameters is a major goal of the Higgs physics program at the LHC. As can be seen in figure 1.3, vertices on these diagrams are labeled with the relevant coupling modifiers. In red are Higgs-quark couplings, κ_t for top quarks and κ_b for bottom quarks. In blue are bosonic couplings, κ_V for a general vector boson, and κ_W for a $W^{\pm}$.

Not all allowed production modes are of equal probability, however; each possible mode has an associated cross-section, represented by σ. This cross-secton is generally measured in a unit of area called a picobarn (pb), where one picobarn is equal to $10^{-36}\ cm^2$. The cross-section

Table 1.1. Table showing the calculated cross-sections (in pb) of the single Higgs boson production modes for $\sqrt{s} = 13$ TeV. This data has been taken from [14].

Production Mode	Cross-section (pb)
ggH	48.31 ± 2.44
VBF	3.771 ± 0.807
WH	1.359 ± 0.028
ZH	0.877 ± 0.036
ttH	0.503 ± 0.035
bbH	0.482 ± 0.097
tH	0.092 ± 0.008

depends on the total center-of-mass collision energy, $\sqrt{s}$ (measured in units of energy, in this case teraelectronvolt, TeV). At a center of mass energy of $\sqrt{s} = 13$ TeV, recent results have shown the total Higgs boson production cross section as $\sigma_H = 54 \pm 2.6$ pb [14]. Gluon fusion production accounts for about 87% of this total cross-section [13]. Taken together with the next most common mode, VBF, over 93% of the total cross-section can be accounted for. The cross-section for different production modes are tabulated in table 1.1. The proportions are represented in the pie chart shown in figure 1.4, reproduced from [13].

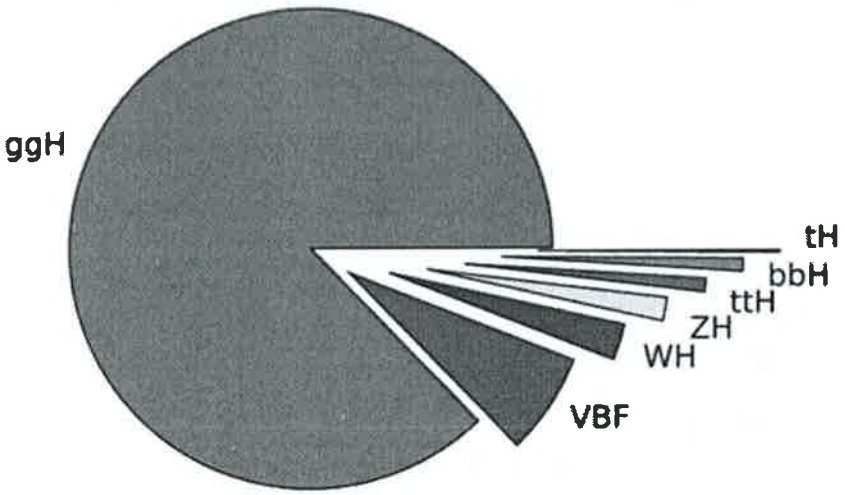


Figure 1.4. A pie chart of the relative proportions of the total σ_H attributed to each production mode. Reproduced from [13].

1.3.2 Decay Modes of a Single Higgs Boson

While section 1.3.1 covers the ways to create a Higgs boson at the LHC, the ways in which the Higgs decays are of even more vital importance. With a SM lifetime of $\tau_H = 1.6 \times 10^{-22}$ s, the Higgs itself cannot be directly measured [13]. Instead, the final state particles must be measured and the Higgs reconstructed from them. Feynman diagrams for the possible decay modes are show in figure 1.5. There are 4 major classes of Higgs decays: vector bosonic decays where the final state is a pair of W or Z bosons (figure 1.5g); fermionic decays where the final state is a fermion-antifermion pair (figure 1.5h); photonic decays where the final state is either two photons or a photon and a Z^0 boson (figure 1.5i&j); and decays into two gluons, which are mediated by a (b or t) quark loop (not shown in figure 1.5).

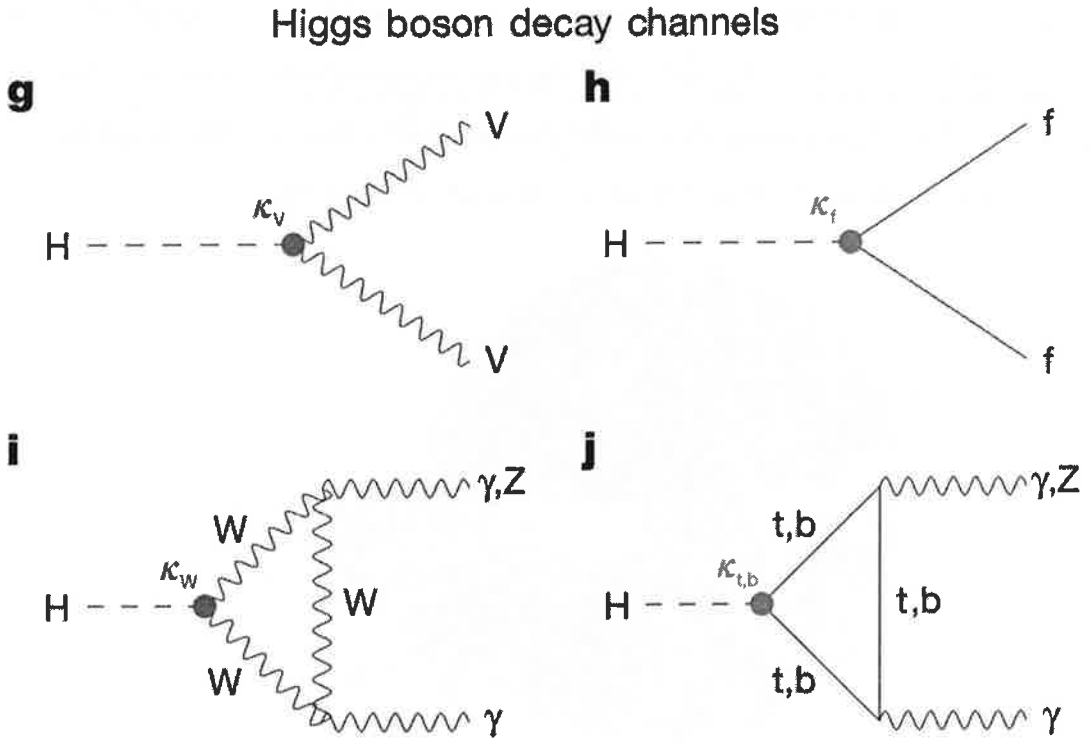


Figure 1.5. Feynman diagrams of known Higgs decay modes. Heavy vector bosonic decays are shown in **g**, fermion-antifermion decays are shown in **h**, and photon/Z boson decays are shown in **i** and **j**. Reproduced from [13]

Table 1.2. Table showing the theoretical branching fractions of various Higgs decay channels at $\sqrt{s} = 13$ TeV. This data has been taken from [14].

Decay Channel	Branching Fraction (%)
$b\bar{b}$	57.63 ± 0.70
WW	22.00 ± 0.33
gg	8.15 ± 0.42
$\tau\tau$	6.21 ± 0.09
cc	2.86 ± 0.09
ZZ	2.71 ± 0.04
$\gamma\gamma$	0.227 ± 0.005
$Z\gamma$	0.157 ± 0.009
ss	0.025 ± 0.001
$\mu\mu$	0.0216 ± 0.0004

As with the production modes, not all of the decay modes are equally likely. Whereas a production mode is assigned a physical cross-section, a decay mode is described by the branching faction, which quantifies what percent of all produced particles will decay through a certain channel. These branching fractions, as with the production cross-sections, can be both theoretically calculated and measured from experiments. Table 1.2 shows the branching fractions of the main Higgs decay modes with information obtained from [14]. The total fractions are also shown as a pie chart in figure 1.6, which has been reproduced from [13].

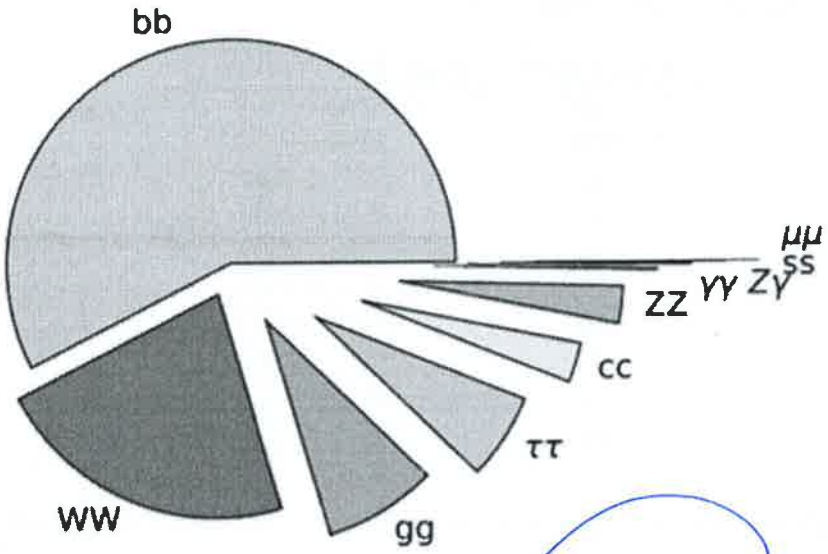


Figure 1.6. Theoretical branching fractions for a single Higgs. Reproduced from [13]

As seen in the table, the $b\bar{b}$ decay channel is by far the most likely for the Higgs boson. This channel therefore will be produced at high rate at the LHC. The b quarks, however, will produce many hadronic jets as described in 1.1.2. So, while this decay is very high-rate, it can be quite hard to measure reliable due to the preponderance of produced particles. The WW (heavy vector boson) decay is the next most common. The W bosons can decay hadronically, semi-leptonically, or fully leptonically. The result, regardless of the decay channel of the W bosons, is a final state with a large number of particles that must be correctly tagged and measured. While this decay mode has been used in previous analyses, it hinges upon accurate tagging of jets and an understanding of the missing transverse momentum carried off by neutrinos produced in the leptonic decay channel. Together, these two high-rate but less clean signal signatures make up more than 75% of the total branching fraction for single Higgs decay.

1.4 Higgs Boson Decaying to Two Photons ($H \rightarrow \gamma\gamma$)

For this analysis, the major decay channel of interest will be the diphoton, or $\gamma\gamma$ decay. This is, again, represented by the Feynman diagrams in figure 1.5i&j. Looking at table 1.2 and figure 1.6, it is clear that this decay channel has a very low branching fraction compared to some of the higher-rate channels at only 0.2%. Its major advantage, however, comes from the purity of signal. As discussed in section 1.1.2, pp scatting processes produce a nearly-uncountable number of QCD jets. So, any final state where jets are expected can be very hard to separate from the backgrounds at play. Additionally, each of these jets can contain dozens to hundreds of individual signatures, and are therefore very challenging to reconstruct with high accuracy.

Diphoton decays sidestep many of these issues with QCD-like decay channels. While there will be many background photons, their rate is considerably lower than that of jets. Additionally, photon signatures are relatively well-resolved and much easier to properly reconstruct than jet signatures. As will be discussed in section 3.2.1, photon signatures tend to be well-columnated and more likely to be isolated from other signatures. Furthermore, since the CMS detector places the detector element that measures photon signatures closer to the particle interaction point, photons tend to spread out less than QCD jets and produce fewer additional particles for each real signature. These factors lead to a much easier-to-measure invariant mass of the two photons compared to that of decays including jets. Overall, the diphoton decay channel provides the best mass resolution, which is a key consideration when making precise measurements of the properties of the Higgs boson. So, then, the $H \rightarrow \gamma\gamma$ has always proven to be a very fruitful Higgs decay despite its low rate.

1.5 Di-Higgs Production and Properties

Another major area of investigation are in diHiggs production modes. One major benefit of such analyses is the ability to probe the Higgs self-coupling, λ, introduced in equation 1.18 [13]. Since both m_H and λ are free parameters in the BEH potential, experimentally constraining

both m_H and λ would allow for a more complete picture of the overall BEH field. This process provides one way to probe possible beyond standard model physics, as a better understanding of the BEH field may point fundamental physics in a new direction.

As with single Higgs physics, DiHiggs processes can be understood in terms of their production and decay modes. There is no additional physics to be found in DiHiggs decays; once the two independent Higgs bosons are produced, they both decay exactly according to the decay modes described in section 1.3.2. The possible Higgs boson pair production modes, however, are of considerable interest. There are three major ways that Higgs boson pairs can be produced: a single Higgs boson can be produced at an interaction vertex, then can split into two Higgs through the trilinear coupling; a pair of Higgs bosons can be produced at a single vertex with a vector boson; two separate vertices in an interaction can produce their own Higgs bosons, which in no way share an interaction point.

1.5.1 Di-Higgs Production Modes

There are two ggH processes, shown in figure 1.7k&l. The first, **k**, shows the creation of a virtual Higgs, H^*, through the same standard top/bottom quark loop as the single Higgs ggH process. This virtual Higgs then decays into two further on-shell (real) Higgs bosons. This process has a tri-linear Higgs coupling vertex, allowing the direct probing of the Higgs self-interaction, κ_λ. The second ggH process does not have this tri-linear coupling, instead having a 4-part virtual top/bottom quark loop where one Higgs is produced at each of the forward corners.

There are three separate VBF modes for DiHiggs production. The first, **m**, involves the fusion of two vector bosons, then the creation of a virtual H^* which then decays into two final state Higgs bosons. As a result, this production moode probes both κ_λ and κ_V. The next diagram, **n**, follows the same process but skips the virtual Higgs. Instead, two Higgs bosons are produced directly at the interaction vertex between the two vector bosons. This introduces a new coupling modifier, κ_{2V}. The final diagram in **o** consists of the exchange of a vector boson, leading to two

individual κ_V vertices.

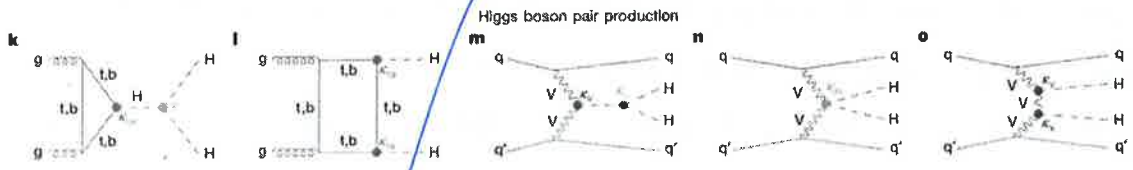


Figure 1.7. Feynman diagrams of the most common theorized DiHiggs production modes. Figures **k** and **l** correspond to the two methods of ggH production, and **m**,**n**, and **o** correspond to the three methods of VBF production. Reproduced from [13].

When calculating the scattering amplitudes for DiHiggs production, an interesting phenomenon occurs. The ggH production modes, which are the two most common, actually produce amplitudes of similar magnitude but opposite signs [13]. So, destructive interference between the amplitudes occurs. This leads to a very small total production cross-section. In fact, whereas the single Higgs production cross section is about $\sigma_H = 54$ pb, the theoretical HH cross-section is calculated to be $\sigma_{HH=32.76^{+1.95}_{-6.83}}$ fb, where a femtobarn is 1/1000th the size of a petabarn. So, then, HH processes are theoretically more than a factor of 1000 more rare than single Higgs processes. As a result, no significant diHiggs signals have been measured in Run 2 or Run 3 of the LHC [13].

1.5.2 The $HH \to b\bar{b}\gamma\gamma$ Channel

Given the incredibly low expected rate of HH production as mentioned in the previous section, all possible pains must be taken to maximize the signal efficiency for relatively high-rate decays of diHiggs production. The $H \to \gamma\gamma$ decay is still useful for single Higgs production – as discussed in section 1.4 – because even though it has a relatively low rate, single Higgs production is common enough that the ease of separating the signal from background is a worthwhile tradeoff. That same logic doesn't hold for HH production however; since 0.2% of single Higgs decay to two photons, only 0.004% of HH productions would decay to four photons.

In order to get around this low rate but still take advantage of the ease of identifying $H \to \gamma\gamma$, the decay $H \to \bar{b}b$ can be used. $H \to \bar{b}b$ has a branching fraction of 57.63%, but can be

quite hard to separate from background given its jet-jet final state. Combining this higher-rate process with the much more easyily disentangled $H \to \gamma\gamma$ process, however, provides the best possible compromise between rate and ease of identification. Additionally, the mass resolution is worth consideration. Though the $b\bar{b}$ channel provides high rate, it often has a much worse mass resolution than the diphoton channel. Therefore, having an easier-to-measure $H \to \gamma\gamma$ decay allows the mass of the first Higgs to be clearly identified and well-tagged. This means that the overall quality of the analysis can be improved despite the lower rate that occurs when requiring a diphoton decay.